

$$- \Delta U = Q + W$$

**15**

**C**

$$\begin{aligned} v &= \omega \sqrt{x_0^2 - x^2} \\ &= (2\pi(1.0))\sqrt{4^2 - 3^2} \\ &= 16.6 \approx 17 \text{ cm s}^{-1} \end{aligned}$$